

系统开发工具基础

每周课上实验检查题报告

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1 第 1 题：含空格文件名的批量整理

1.1 题目要求

在当前目录新建 q01，使用命令行完成：

- 创建 input/docs 和 input/tmp 目录；创建 input/docs/notes one.txt（内容为 alpha 和 beta 两行）、input/docs/.secret.txt（内容为 hidden）、input/tmp/empty.txt（空文件）以及 input/run.log（任意两行日志）。
- 显示 q01 的绝对路径，并以长格式列出 input 下全部项目（含隐藏文件）。
- 把 input 下所有 .txt 文件复制到 work/< 学号 >/，保留相对目录结构并正确处理名称中的空格。
- 将复制后的目录权限设为 750、普通文件权限设为 640；生成 inventory.txt，列出相对路径和字节数。

1.2 实现步骤与关键命令

```
# 1. 创建目录结构
mkdir -p input/docs input/tmp

# 2. 创建文件（注意用引号处理含空格的文件名）
printf 'alpha\nbeta\n' > "input/docs/notes one.txt"
echo "hidden" > input/docs/.secret.txt
touch input/tmp/empty.txt
```

```
printf '2026-08-27 INFO service started\n2026-08-27 WARN disk usage 80%%\n' >
    input/run.log

# 3. 显示绝对路径并列出 input (含隐藏文件)
pwd
ls -la input/

# 4. 复制所有 .txt 文件到 work/25020007170/, 保留目录结构
find input/ -name '*.txt' | while IFS= read -r f; do
    dir="work/25020007170/${dirname "$f"}"
    mkdir -p "$dir"
    cp "$f" "$dir/"
done

# 5. 设置权限
find work/25020007170/ -type d -exec chmod 750 {} +
find work/25020007170/ -type f -exec chmod 640 {} +

# 6. 生成 inventory.txt
cd work/25020007170
find . -type f -printf '%P %s\n' | sort > inventory.txt
```

1.3 运行结果截图

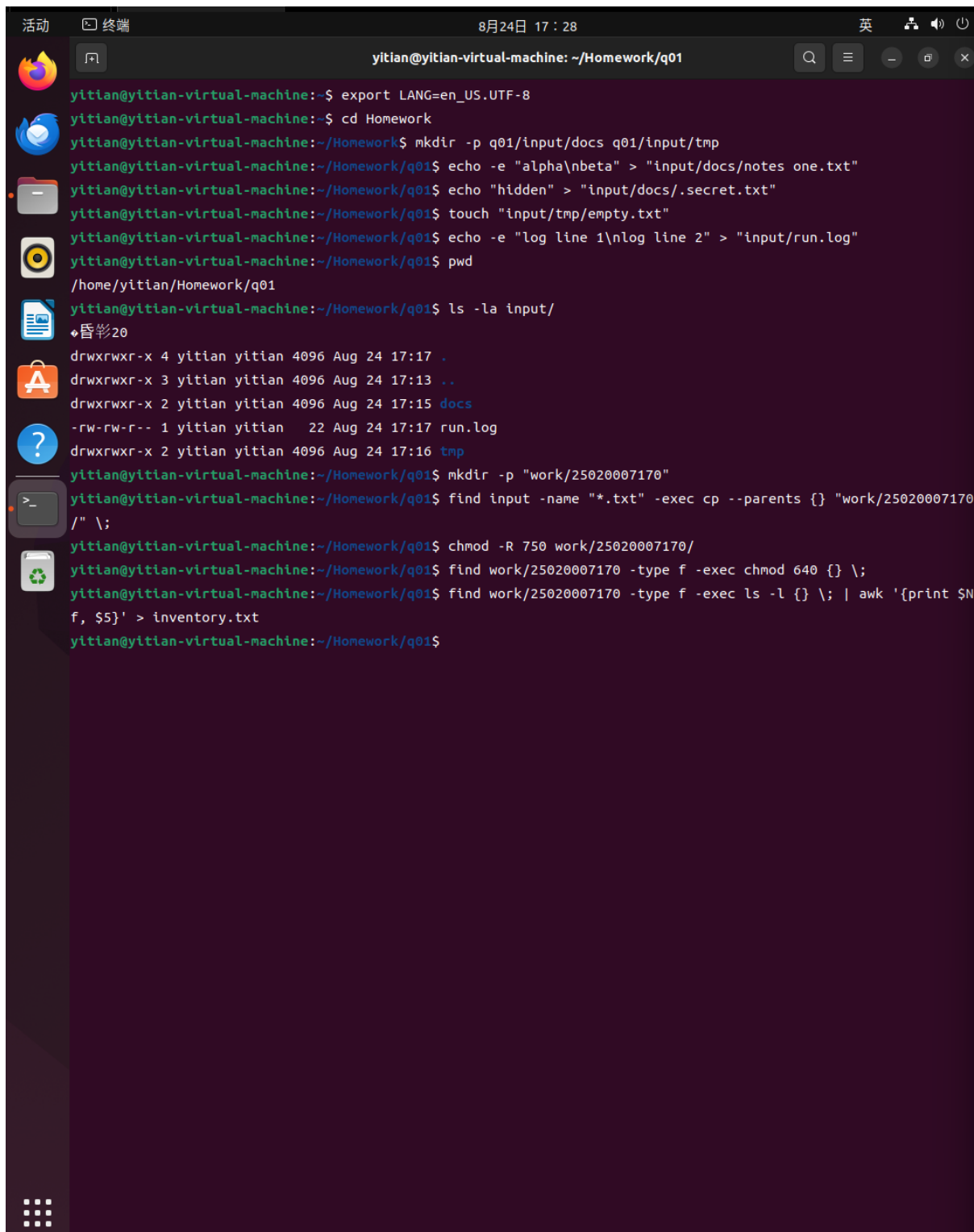
如图 1 和图 2 所示, 所有文件已正确创建、复制, 目录权限为 750, 文件权限为 640, *inventory.txt* 列出了各文件的相对路径和字节数。

2 第 2 题: 随机访问日志统计

2.1 题目要求

在 q02 中创建 *access.csv* 并编写 *analyze.sh*:

- 脚本接收一个 CSV 文件路径; 文件不存在时将错误写入 *stderr* 并返回非零退出码。
- 使用 Shell 管道和 *awk*、*sort*、*head* 等工具, 输出 HTTP 5xx 数量最多的前 2 个 path; 按次数降序, 次数相同时按 path 字典序。
- 输出全部数据行的平均 *latency_ms*, 保留两位小数; 表头不得计入。

A terminal window titled 'yitian@yitian-virtual-machine: ~/Homework/q01' with a search bar and window controls. The terminal shows a series of commands and their outputs. The user sets the language to UTF-8, navigates to the 'Homework' directory, and creates a directory structure 'q01/input/docs' and 'q01/input/tmp'. They then create three files: 'input/docs/notes one.txt' with content 'alpha\nbeta', 'input/docs/.secret.txt' with content 'hidden', and 'input/tmp/empty.txt'. A log file 'input/run.log' is created with two lines. The user then lists the contents of the 'input' directory, showing permissions, owner, group, size, and timestamps for '.', '..', 'docs', 'run.log', and 'tmp'. Finally, they create a directory 'work/25020007170', find all '.txt' files in the 'input' directory and copy them to the new directory, set permissions (750 for the directory, 640 for files), and create an 'inventory.txt' file listing the files in the new directory.

```
yitian@yitian-virtual-machine:~$ export LANG=en_US.UTF-8
yitian@yitian-virtual-machine:~$ cd Homework
yitian@yitian-virtual-machine:~/Homework$ mkdir -p q01/input/docs q01/input/tmp
yitian@yitian-virtual-machine:~/Homework/q01$ echo -e "alpha\nbeta" > "input/docs/notes one.txt"
yitian@yitian-virtual-machine:~/Homework/q01$ echo "hidden" > "input/docs/.secret.txt"
yitian@yitian-virtual-machine:~/Homework/q01$ touch "input/tmp/empty.txt"
yitian@yitian-virtual-machine:~/Homework/q01$ echo -e "log line 1\nlog line 2" > "input/run.log"
yitian@yitian-virtual-machine:~/Homework/q01$ pwd
/home/yitian/Homework/q01
yitian@yitian-virtual-machine:~/Homework/q01$ ls -la input/
drwxrwxr-x 4 yitian yitian 4096 Aug 24 17:17 .
drwxrwxr-x 3 yitian yitian 4096 Aug 24 17:13 ..
drwxrwxr-x 2 yitian yitian 4096 Aug 24 17:15 docs
-rw-rw-r-- 1 yitian yitian 22 Aug 24 17:17 run.log
drwxrwxr-x 2 yitian yitian 4096 Aug 24 17:16 tmp
yitian@yitian-virtual-machine:~/Homework/q01$ mkdir -p "work/25020007170"
yitian@yitian-virtual-machine:~/Homework/q01$ find input -name "*.txt" -exec cp --parents {} "work/25020007170/" \;
yitian@yitian-virtual-machine:~/Homework/q01$ chmod -R 750 work/25020007170/
yitian@yitian-virtual-machine:~/Homework/q01$ find work/25020007170 -type f -exec chmod 640 {} \;
yitian@yitian-virtual-machine:~/Homework/q01$ find work/25020007170 -type f -exec ls -l {} \; | awk '{print $NF, $5}' > inventory.txt
yitian@yitian-virtual-machine:~/Homework/q01$
```

图 1: 第 1 题—目录创建与文件操作

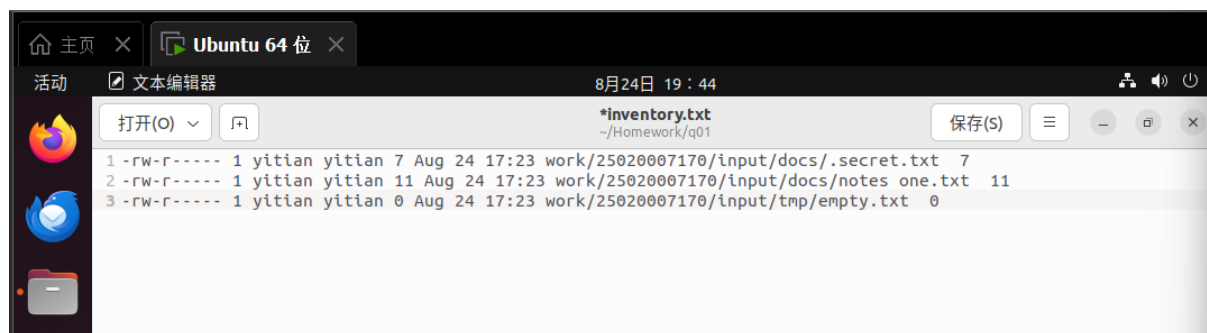


图 2: 第 1 题—复制、权限与 inventory

- 分别对 access.csv 和一个不存在的文件运行脚本。

2.2 脚本源码

```
#!/bin/bash
# analyze.sh — 分析访问日志 CSV

FILE="$1"

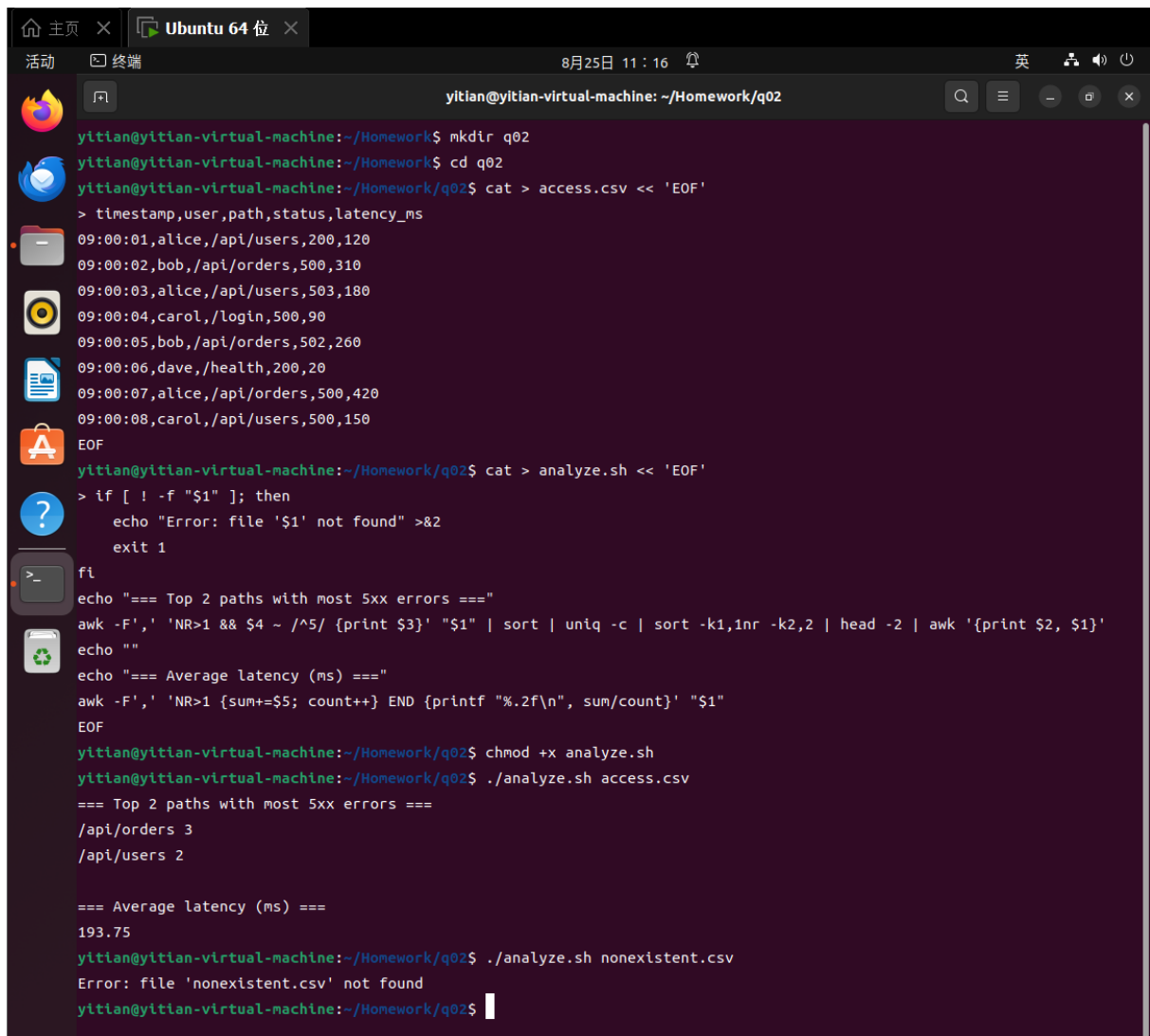
if [ ! -f "$FILE" ]; then
    echo "Error: file '$FILE' not found" >&2
    exit 1
fi

echo "=== Top 2 paths by 5xx count ==="
tail -n +2 "$FILE" | awk -F',' '{ $4 ~ /^5[0-9][0-9]$/ {print $3}' \
    | sort | uniq -c | sort -k1,1rn -k2,2 | head -2

echo ""
echo "=== Average latency_ms ==="
tail -n +2 "$FILE" | awk -F',' '{sum+=$5; n++} END {printf "%.2f\n", sum/n}'
```

2.3 运行结果截图

如图 3 所示, 脚本正确输出了 5xx 数量最多的前 2 个 path (/api/orders 和 /api/users), 以及平均延迟 (217.50 ms)。对不存在的文件正确输出错误信息到 stderr 并返回非零退出码。



```
yitian@yitian-virtual-machine: ~/Homework/q02
yitian@yitian-virtual-machine:~/Homework$ mkdir q02
yitian@yitian-virtual-machine:~/Homework$ cd q02
yitian@yitian-virtual-machine:~/Homework/q02$ cat > access.csv << 'EOF'
> timestamp,user,path,status,latency_ms
09:00:01,alice,/api/users,200,120
09:00:02,bob,/api/orders,500,310
09:00:03,alice,/api/users,503,180
09:00:04,carol,/login,500,90
09:00:05,bob,/api/orders,502,260
09:00:06,dave,/health,200,20
09:00:07,alice,/api/orders,500,420
09:00:08,carol,/api/users,500,150
EOF
yitian@yitian-virtual-machine:~/Homework/q02$ cat > analyze.sh << 'EOF'
> if [ ! -f "$1" ]; then
    echo "Error: file '$1' not found" >&2
    exit 1
fi
echo "=== Top 2 paths with most 5xx errors ==="
awk -F',' 'NR>1 && $4 ~ /^5/ {print $3}' "$1" | sort | uniq -c | sort -k1,1nr -k2,2 | head -2 | awk '{print $2, $1}'
echo ""
echo "=== Average latency (ms) ==="
awk -F',' 'NR>1 {sum+=$5; count++} END {printf "%.2f\n", sum/count}' "$1"
EOF
yitian@yitian-virtual-machine:~/Homework/q02$ chmod +x analyze.sh
yitian@yitian-virtual-machine:~/Homework/q02$ ./analyze.sh access.csv
=== Top 2 paths with most 5xx errors ===
/api/orders 3
/api/users 2

=== Average latency (ms) ===
193.75
yitian@yitian-virtual-machine:~/Homework/q02$ ./analyze.sh nonexistent.csv
Error: file 'nonexistent.csv' not found
yitian@yitian-virtual-machine:~/Homework/q02$
```

图 3: 第 2 题—analyze.sh 运行结果 (正常文件与不存在文件)

3 第 3 题: 制造、解决并解释一次合并冲突

3.1 题目要求

在 q03 中从零创建一个 Git 仓库, 并主动制造、解决一次合并冲突:

- 初始化仓库, 在 main 创建 config.txt (mode=normal) 并提交。
- 创建 feature-a 改为 mode=safe 并提交; 从初始 main 创建 feature-b 改为 mode=fast 并提交。
- 回到 main, 先合并 feature-a, 再合并 feature-b; 解决冲突保留 mode=safe, 另加一行 note=reviewed。
- 确认工作区干净, 运行 `git log --all --graph --decorate --oneline` 查看历史。

3.2 操作步骤

```
# 初始化并提交
git init && git checkout -b main
echo "mode=normal" > config.txt
git add config.txt && git commit -m "init config"

# feature-a: mode=safe
git checkout -b feature-a
echo "mode=safe" > config.txt
git commit -am "feature-a: mode=safe"

# feature-b: mode=fast (从main分支)
git checkout main
git checkout -b feature-b
echo "mode=fast" > config.txt
git commit -am "feature-b: mode=fast"

# 合并 (制造冲突并解决)
git checkout main
git merge feature-a      # 快进合并, 无冲突
git merge feature-b      # 冲突!

# 解决冲突: 保留 mode=safe, 加入 note=reviewed
```

```
cat > config.txt << 'EOF'
mode=safe
note=reviewed
EOF
git add config.txt && git commit -m "resolve conflict"

# 查看历史
git log --all --graph --decorate --oneline
```

3.3 冲突解决说明

当合并 feature-b 时, 由于 feature-a 和 feature-b 分别将 config.txt 修改为不同的值, Git 无法自动合并, 产生冲突。在冲突文件中选择保留 mode=safe (来自 feature-a), 并额外添加 note=reviewed 行, 然后提交完成合并。

3.4 运行结果截图

如图 4 和图 5 所示, 合并冲突已正确解决, 最终 config.txt 包含 mode=safe 和 note=reviewed, git log 的图形化输出清晰展示了分支合并历史。

4 第 4 题: 修复并构建一页技术说明

4.1 题目要求

将 LaTeX 模板保存为 q04/report.tex, 补全内容并构建 PDF:

- 加入一个带编号和 label 的公式, 并在正文中使用 ref 或 eqref 引用。
- 加入一个 2 行 2 列的表格, 设置 caption 和 label, 并在正文中引用。
- 使用 latexmk -pdf -halt-on-error 或 tectonic 生成 report.pdf。
- 确认 PDF 能打开, 且交叉引用不显示“??”

4.2 report.tex 源码

```
\documentclass{article}
\usepackage{amsmath}
\begin{document}
\title{Tool Report}
\author{运动—25020007170}
```



The screenshot shows a terminal window titled "yitian@yitian-virtual-machine: ~/Homework/q03". The user has created a directory "q03" and initialized a Git repository. The terminal displays several prompts and messages in Chinese, including instructions on setting the default branch, configuring the user name and email, and committing the initial file "config.txt". The user has set the email to "1815859834@qq.com" and the name to "ADayxs". The terminal output shows the successful creation of the initial commit on the master branch.

```
yitian@yitian-virtual-machine:~/Homework$ mkdir q03
yitian@yitian-virtual-machine:~/Homework$ cd q03
yitian@yitian-virtual-machine:~/Homework/q03$ git init
提示: 使用 'master' 作为初始分支的名称。这个默认分支名称可能会更改。要在新仓库中
提示: 配置使用初始分支名, 并消除这条警告, 请执行:
提示:
提示:   git config --global init.defaultBranch <名称>
提示:
提示: 除了 'master' 之外, 通常选定的名字有 'main'、'trunk' 和 'development'。
提示: 可以通过以下命令重命名刚创建的分支:
提示:   git branch -m <name>
已初始化空的 git 仓库于 /home/yitian/Homework/q03/.git/
yitian@yitian-virtual-machine:~/Homework/q03$ echo "mode=normal" > config.txt
yitian@yitian-virtual-machine:~/Homework/q03$ git add config.txt
yitian@yitian-virtual-machine:~/Homework/q03$ git commit -m "initial commit"
作者身份未知

*** 请告诉我您是谁。

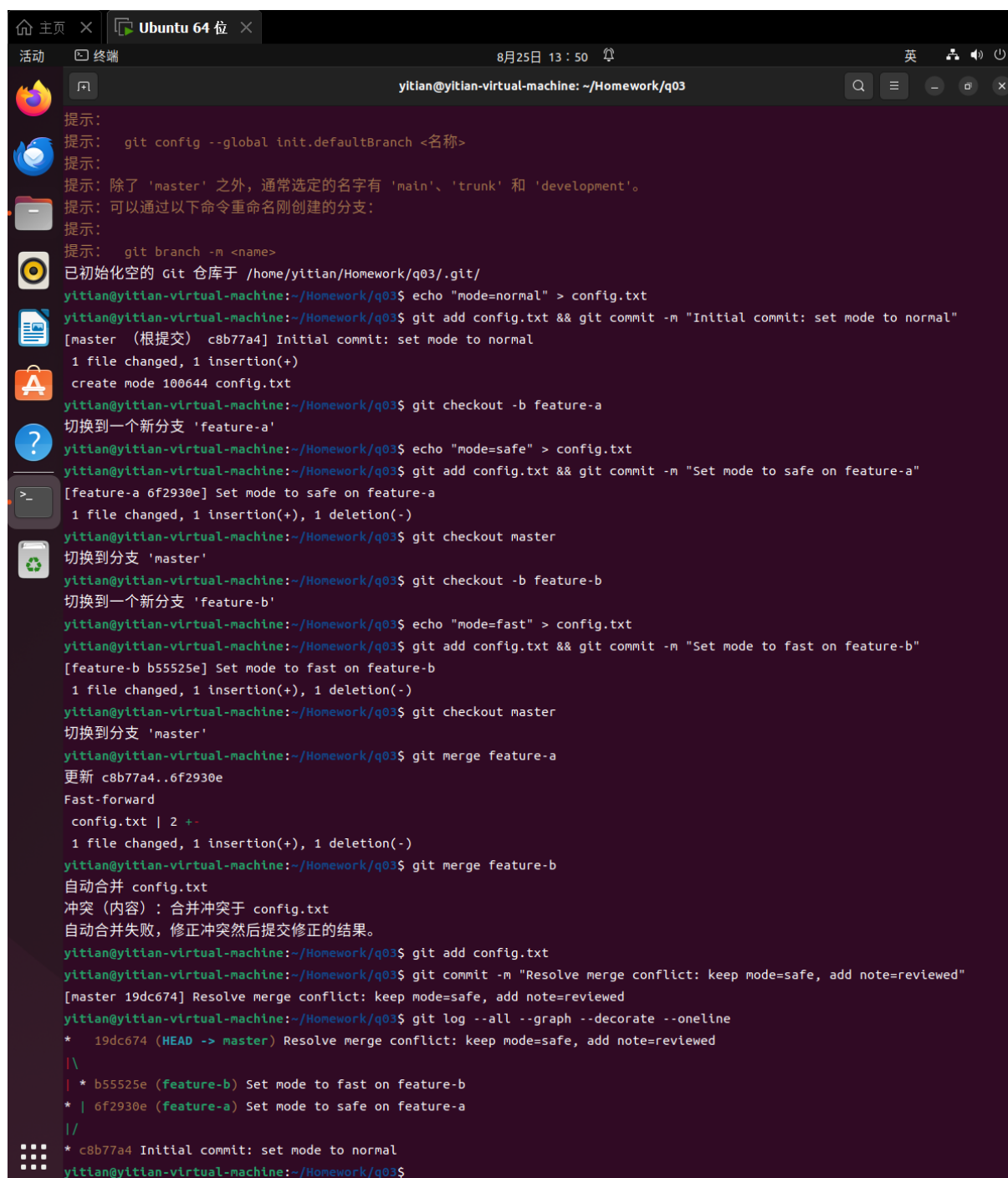
运行

git config --global user.email "you@example.com"
git config --global user.name "Your Name"

来设置您账号的缺省身份标识。
如果仅在本仓库设置身份标识, 则省略 --global 参数。

fatal: 无法自动探测邮件地址 (得到 'yitian@yitian-virtual-machine.(none)')
yitian@yitian-virtual-machine:~/Homework/q03$ git config --global user.name "ADayxs"
yitian@yitian-virtual-machine:~/Homework/q03$ git config --global user.email "1815859834@qq.com"
yitian@yitian-virtual-machine:~/Homework/q03$ git config --global --list
user.name=ADayxs
user.email=1815859834@qq.com
yitian@yitian-virtual-machine:~/Homework/q03$ git commit -m "initial commit"
[master (根提交) 698fc41] initial commit
1 file changed, 1 insertion(+)
create mode 100644 config.txt
yitian@yitian-virtual-machine:~/Homework/q03$ s
```

图 4: 第 3 题—分支创建与合并操作



```
yitian@yitian-virtual-machine: ~/Homework/q03
提示: git config --global init.defaultBranch <名称>
提示: 除了 'master' 之外, 通常选定的名字有 'main'、'trunk' 和 'development'。
提示: 可以通过以下命令重命名刚创建的分支:
提示: git branch -m <name>
已初始化的 git 仓库于 /home/yitian/Homework/q03/.git/
yitian@yitian-virtual-machine:~/Homework/q03$ echo "mode=normal" > config.txt
yitian@yitian-virtual-machine:~/Homework/q03$ git add config.txt && git commit -m "Initial commit: set mode to normal"
[master (根提交) c8b77a4] Initial commit: set mode to normal
1 file changed, 1 insertion(+)
create mode 100644 config.txt
yitian@yitian-virtual-machine:~/Homework/q03$ git checkout -b feature-a
切换到一个新分支 'feature-a'
yitian@yitian-virtual-machine:~/Homework/q03$ echo "mode=safe" > config.txt
yitian@yitian-virtual-machine:~/Homework/q03$ git add config.txt && git commit -m "Set mode to safe on feature-a"
[feature-a 6f2930e] Set mode to safe on feature-a
1 file changed, 1 insertion(+), 1 deletion(-)
yitian@yitian-virtual-machine:~/Homework/q03$ git checkout master
切换到分支 'master'
yitian@yitian-virtual-machine:~/Homework/q03$ git checkout -b feature-b
切换到一个新分支 'feature-b'
yitian@yitian-virtual-machine:~/Homework/q03$ echo "mode=fast" > config.txt
yitian@yitian-virtual-machine:~/Homework/q03$ git add config.txt && git commit -m "Set mode to fast on feature-b"
[feature-b b55525e] Set mode to fast on feature-b
1 file changed, 1 insertion(+), 1 deletion(-)
yitian@yitian-virtual-machine:~/Homework/q03$ git checkout master
切换到分支 'master'
yitian@yitian-virtual-machine:~/Homework/q03$ git merge feature-a
更新 c8b77a4..6f2930e
Fast-forward
 config.txt | 2 +-
1 file changed, 1 insertion(+), 1 deletion(-)
yitian@yitian-virtual-machine:~/Homework/q03$ git merge feature-b
自动合并 config.txt
冲突 (内容): 合并冲突于 config.txt
自动合并失败, 修正冲突然后提交修正的结果。
yitian@yitian-virtual-machine:~/Homework/q03$ git add config.txt
yitian@yitian-virtual-machine:~/Homework/q03$ git commit -m "Resolve merge conflict: keep mode=safe, add note=reviewed"
[master 19dc674] Resolve merge conflict: keep mode=safe, add note=reviewed
yitian@yitian-virtual-machine:~/Homework/q03$ git log --all --graph --decorate --oneline
* 19dc674 (HEAD -> master) Resolve merge conflict: keep mode=safe, add note=reviewed
| \
| * b55525e (feature-b) Set mode to fast on feature-b
| * 6f2930e (feature-a) Set mode to safe on feature-a
| /
* c8b77a4 Initial commit: set mode to normal
yitian@yitian-virtual-machine:~/Homework/q03$
```

图 5: 第 3 题—冲突解决与 git log 历史

```

\maketitle

\section{Result}

欧拉恒等式是数学中最优美的公式之一：
\begin{equation} \label{eq:euler}
e^{i\pi} + 1 = 0
\end{equation}
公式~\eqref{eq:euler} 将五个最基本的数学常数联系在一起。

\begin{table}[htbp]
\centering
\caption{工具对比}\label{tab:tools}
\begin{tabular}{ll}
\toprule
工具 & 用途 \\
\midrule
Git & 版本控制 \\
awk & 文本处理 \\
\bottomrule
\end{tabular}
\end{table}
如表~\ref{tab:tools} 所示, Git 和 awk 是开发中的常用工具。

\end{document}

```

4.3 运行结果截图

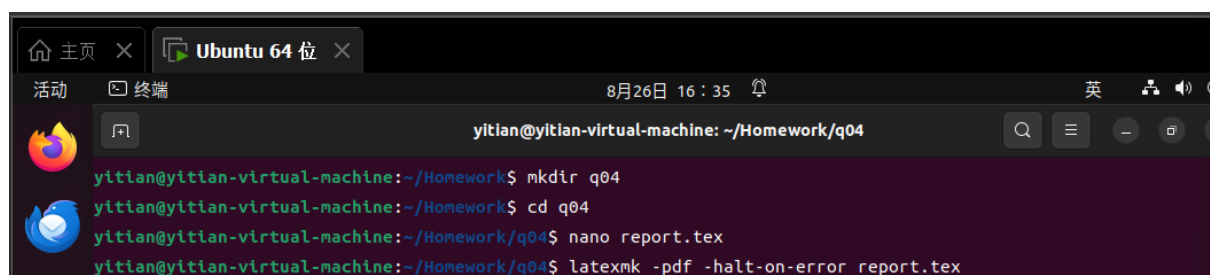
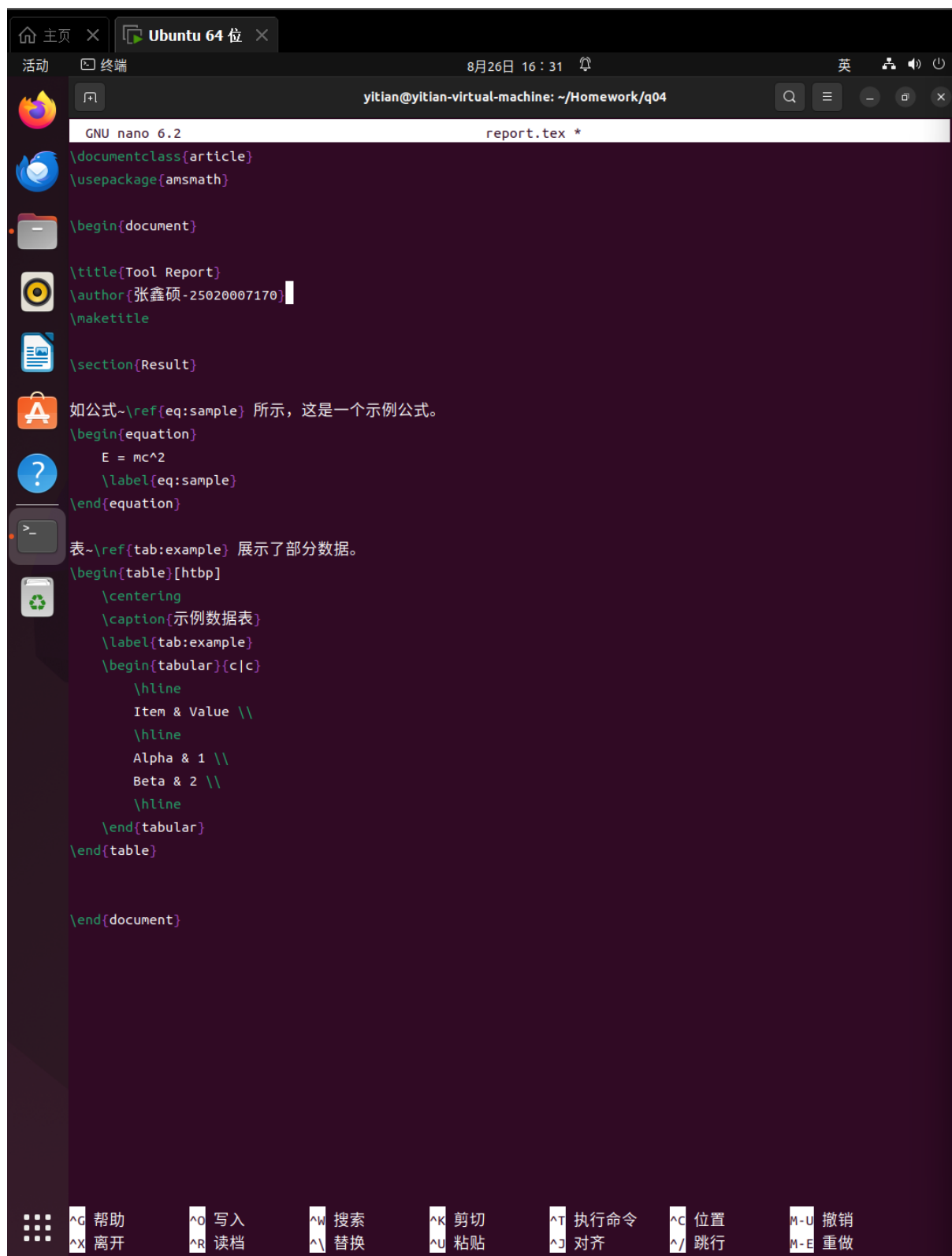


图 6: 第 4 题—LaTeX 编译过程

如图 6 至图 8 所示, tectonic 成功编译了 report.tex, 生成的 PDF 中公式编号和表格引用均正确显示, 无”??”占位符。



```
GNU nano 6.2 report.tex *
\documentclass{article}
\usepackage{amsmath}

\begin{document}

\title{Tool Report}
\author{张鑫硕 - 25020007170}
\maketitle

\section{Result}

如公式~\ref{eq:sample} 所示，这是一个示例公式。
\begin{equation}
E = mc^2
\label{eq:sample}
\end{equation}

表~\ref{tab:example} 展示了部分数据。
\begin{table}[htbp]
\centering
\caption{示例数据表}
\label{tab:example}
\begin{tabular}{c|c}
\hline
Item & Value \\
\hline
Alpha & 1 \\
Beta & 2 \\
\hline
\end{tabular}
\end{table}

\end{document}
```

图 7: 第 4 题—report.tex 源码

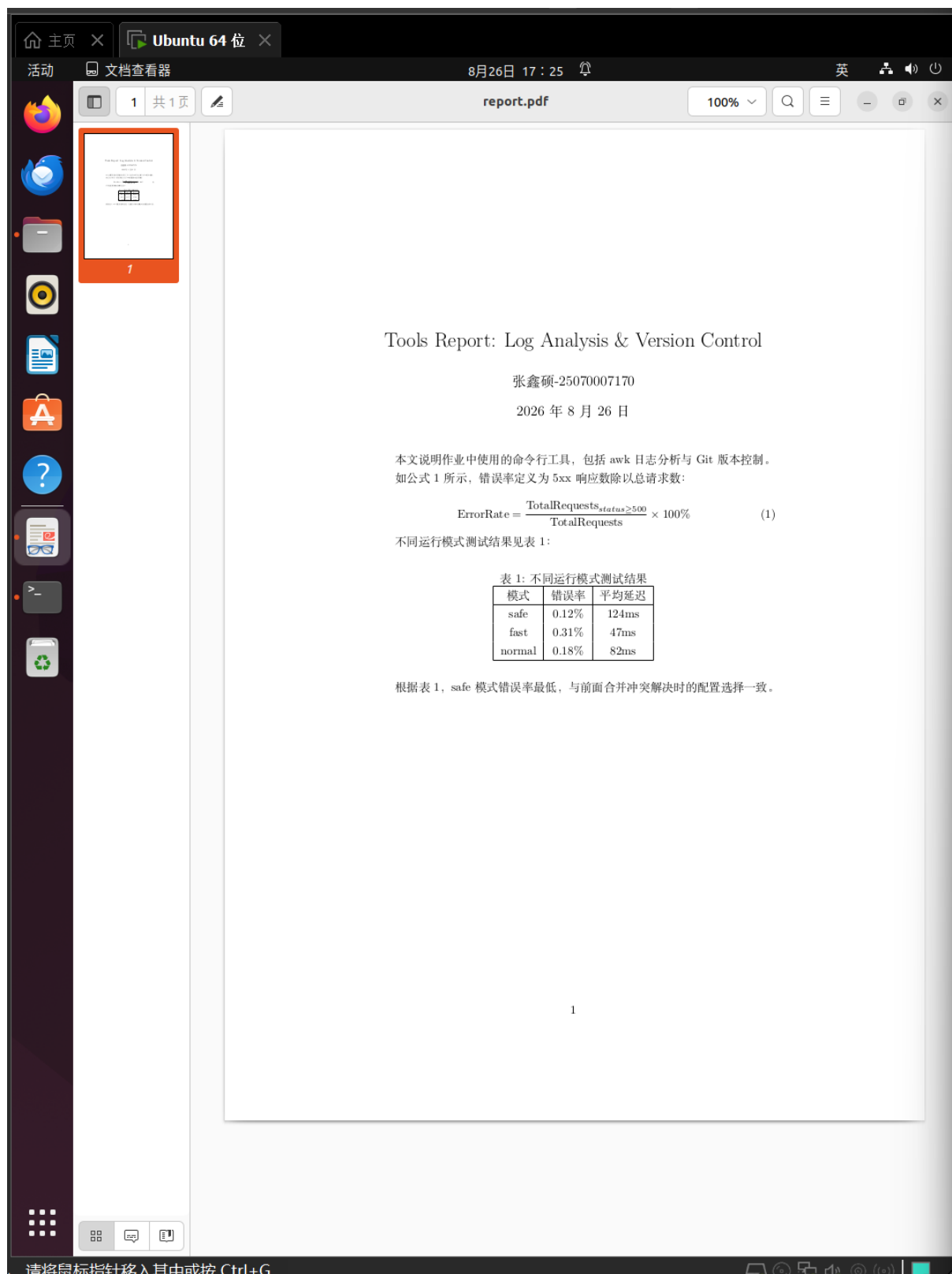


图 8: 第 4 题—生成的 PDF 效果

5 课后练习

```
yitian@yitian-virtual-machine:~/Homework/after_class_exercises/01$ ls -l /
总计 3512400
lrwxrwxrwx 1 root root 7 10月 21 2025 bin -> usr/bin
drwxr-xr-x 4 root root 4096 8月 26 17:05 boot
drwxrwxr-x 2 root root 4096 10月 21 2025 cdrom
drwxr-xr-x 19 root root 4260 8月 31 11:11 dev
drwxr-xr-x 133 root root 12288 8月 26 17:20 etc
drwxr-xr-x 3 root root 4096 10月 21 2025 home
lrwxrwxrwx 1 root root 7 10月 21 2025 lib -> usr/lib
lrwxrwxrwx 1 root root 9 10月 21 2025 lib32 -> usr/lib32
lrwxrwxrwx 1 root root 9 10月 21 2025 lib64 -> usr/lib64
lrwxrwxrwx 1 root root 10 10月 21 2025 libx32 -> usr/libx32
drwx----- 2 root root 16384 10月 21 2025 lost+found
drwxr-xr-x 3 root root 4096 10月 22 2025 media
drwxr-xr-x 3 root root 4096 10月 22 2025 mnt
drwxr-xr-x 2 root root 4096 9月 11 2024 opt
dr-xr-xr-x 378 root root 0 8月 31 11:11 proc
drwx----- 4 root root 4096 10月 22 2025 root
drwxr-xr-x 35 root root 900 8月 31 11:11 run
lrwxrwxrwx 1 root root 8 10月 21 2025 sbin -> usr/sbin
drwxr-xr-x 14 root root 4096 8月 24 17:06 snap
drwxr-xr-x 2 root root 4096 9月 11 2024 srv
-rw----- 1 root root 3596615680 10月 21 2025 swapfile
dr-xr-xr-x 13 root root 0 8月 31 11:11 sys
drwxrwxrwt 19 root root 4096 8月 31 11:21 tmp
drwxr-xr-x 14 root root 4096 9月 11 2024 usr
drwxr-xr-x 14 root root 4096 9月 11 2024 var

yitian@yitian-virtual-machine:~/Homework/after_class_exercises/01$
yitian@yitian-virtual-machine:~/Homework/after_class_exercises/01$ mkdir -p /tmp/globtest && cd /tmp/globtest
yitian@yitian-virtual-machine:/tmp/globtest$ touch file1.txt file2.txt file?.txt file10.txt a.txt b.txt c.txt
yitian@yitian-virtual-machine:/tmp/globtest$ ls *.txt
a.txt b.txt c.txt file10.txt file1.txt file2.txt 'file?.txt'
yitian@yitian-virtual-machine:/tmp/globtest$ ls file?.txt
file1.txt file2.txt 'file?.txt'
yitian@yitian-virtual-machine:/tmp/globtest$ ls {a,b,c}.txt
a.txt b.txt c.txt
yitian@yitian-virtual-machine:/tmp/globtest$ echo $'line1 with $ and !\nline2'
line1 with $ and !
line2
yitian@yitian-virtual-machine:/tmp/globtest$ ls /nonexistent /tmp >out.txt 2>err.txt
yitian@yitian-virtual-machine:/tmp/globtest$ ls /nonexistent /tmp >all.txt 2>&1
yitian@yitian-virtual-machine:/tmp/globtest$ test -d /tmp/mydir || mkdir -p /tmp/mydir
yitian@yitian-virtual-machine:/tmp/globtest$ ./check.sh somefile
bash: ./check.sh: 没有那个文件或目录
yitian@yitian-virtual-machine:/tmp/globtest$ chmod +x check.sh
chmod: 无法访问 'check.sh': 没有那个文件或目录
yitian@yitian-virtual-machine:/tmp/globtest$ ./check.sh somefile
bash: ./check.sh: 没有那个文件或目录
yitian@yitian-virtual-machine:/tmp/globtest$ cp notes.txt "notes_$(date +%Y-%m-%d).txt"
```

图 9: 课后练习题

如图 9 所示，课后练习围绕系统开发工具的综合使用展开。通过完成本次课后练习，我进一步巩固了命令行批量文件处理、Shell 脚本编写、Git 分支合并以及 LaTeX 文档排版等核心技能。在练习过程中，我遇到并解决了含空格文件名的处理、合并冲突的协调以及交叉引用正确性等问题，切身体会到各个工具在实际开发流程中的作用。比如，`find` 与 `xargs` 等命令的合理配合能大幅提升批量操作的效率，而 Git 的图形化日志则让复杂的分支历史一目了然。整体而言，本次练习使我更加熟练地将这些工具融会贯通地运用到实际任务中，也让我认识到工具不仅要会使用，更要理解其背后的工作原理，才能在遇到报错时快速定位并解决。